

**MTM-TAYLOR VERSION 2**  
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**1. INTRODUCTION**

The MTM-Taylor software package permits to simulate deformation textures of polycrystalline materials with f.c.c. and b.c.c. crystal structures. This text is not a complete manual. It will describe:

- how to load the software
- how to run an example of an application.

The application consists of:

- 1) converting an O.D.F. generated by the software system "MTMFHM" into a discrete set of orientations;
- 2) simulation of a plastic deformation by means of the program TAYLORS, using the set of orientations mentioned above as initial orientations;
- 3) converting the resulting sets of orientations (which represent deformation textures) in sets of C-coefficients that can be used by the software system "MTMFHM"

All computer programs run on an IBM-compatible PC, preferably with DOS 5.0. A hard disk (with ca. 3 MBytes of free space) and a mathematical co-processor are needed. The applications 1) and 3) are only possible if the software system MTMFHM is installed on the computer. The programs that interact with MTMFHM-files are all written in Fortran. The simulation programs themselves (application 2) have been written in Pascal (Turbo Pascal Version 5).

The simulation program TAYLORS implements the full constraints and the relaxed constraints Taylor theory as explained in ref [1]. Regarding the choice of optimal solutions, it uses the ideas described in section 6.5 of this paper. The program TAYLORW is similar, but uses the ideas of section 6.4.

## 2. LOADING THE SOFTWARE

The software comes on a floppy disk subdivided in several directories.

- 1) Make a subdirectory on your hard disk with the name "PAS". (It will contain the Pascal-written software).
- 2) Load all files in the root directory of the floppy into this subdirectory "PAS".
- 3) Check the ODF directory (of the MTMFHM software system) on your computer to see whether it contains files named C.001, C.002, C.003, C.004 or ODF.001 that you do not want to lose. If so, make a copy of them to another directory or a floppy disk.
- 4) Load all files of the subdirectory "ODF" of the floppy in the subdirectory "ODF" of your MTMFHM-software system.
- 5) Repeat step 4) for the subdirectories "TEXTURE", "OBJ", "ODFEXE".

## 3. DESCRIPTION OF THE APPLICATION

The manual which follows has been conceived as a detailed description of an exercise. The parameter-files, BAT-files etc. on the diskettes with the software are not identical to those given below as examples.

### 3.1 Converting a measured ODF into a discrete set of crystallite orientations.

A file AODF.001 (generated by MTMFHM) has been copied to the ODF-directory. It contains the values of an C.O.D.F. calculated on a grid in Euler space. Fig. 1 shows this O.D.F. The program ODFTAY has converted it to a set of 399 discrete orientations. It uses the "statistical method" (STAT) described in ref. [2] for this. The following command was issued (from the directory ODF):

```
odftay1
```

This started the procedure ODFTAY1.BAT which looks as follows:

```
rem TO PREPARE A TAYLOR-TEXTURE DATA SET FROM AN ODF
odftay odftay.i01 odftay.l01 odftay.smt odftay.tx0 aodf.001
```

*Explanation:*

- "odftay.i01" is a parameter file. It contains the number of "Selectors" (400 in this case). The "Selectors" are numbers between 0 and 1 that pick up certain crystallite orientations using a sort of "cumulative O.D.F.". The number of selectors should not exceed 5000. The number of crystal orientations that will be generated will be equal or smaller than the number of selectors.
- "odftay.l01" is an output listing.
- "odftay.smt" is an output file that is not used further in this application.
- "odftay.tx0" is the output file that contains the desired set of discrete orientations.
- "aodf.001" is the input file with the O.D.F.

After this run, the file "odftay.tx0" has been copied into the PAS directory.

### 3.2 Simulation of a large plastic deformation

The program TAYLORS was then used to simulate a large plastic deformation, using the texture represented by the orientations in "odftay.tx0" as initial texture. The following command was issued (from the directory PAS):

taylors

The program started and began to read a "control file" "taylors.ctl". This is a fixed name. The file was prepared beforehand. The version used looked as follows, except for the line numbers in italics, which have been added here in order to make the reading of this text easier:

```

1    Put name of file that contains crystal data on next line:
2    fcc.dat
3    1 1 0      LISTING CONTROL PARAMETERS
4    0 0 0.0 0.0  CRSS VALUES (END)
5    0          CRSS-SWITCH
6    0.01       STRAIN RATE EXPONENT M
7    1.000      REFERENCE SLIP RATE GAMMA_DOT_0
8    0          TYPE OF INPUT-TEXTURE-Put filename on next line:
9    odftay.tx0
10   001        CODE TO BE PUT IN OUTPUT-FILE
11   1          NUMBER OF RUNS TO BE PERFORMED
12   0          IF=1:Put name of file with distortions on next line
13   1          FILE CONTROL (IF =1 : Put filename on next line)

```

```

14  tays.tx1
15  FCC roll TAY_S           :Title of this run
16  1      INPUT TEXTURE FILE: CHOSEN RUN
17  0      CHOSEN STEP
18  10     NUMBER OF STEPS TO BE PERFORMED
19  1      NUMBER OF RELAXATIONS
20  0.0 0.0 1.0 RELAXATION 1
21  0.0 0.0 0.0
22  0.0 0.0 0.0
23  0.05   DELTA EPSILON(if zero:only Taylor factor calculation)
24  1.0 0.0 0.0 PRESCRIBED DEFORM.
25  0.0 0.0 0.0
26  0.0 0.0 -1.0

```

*Explanation:*

Since the program was written in Pascal, different rules govern the format of input data as when compared to Fortran programs. The fields in which numerical data are to be given have no fixed length; they must however be separated by spaces. A *real* should look as follows: "3.0" and not as: "3." (write at least one zero after the decimal point). The control file contains some fields with self-explaining comments that are not read by the program. This is the case for the whole of line 1.

line 2: name of the input file that describes the nature of the slip systems. "fcc.dat" stands for {111}<110> slip systems. "fcct.dat" stands for {111}<110> slip systems + {111}<112> mechanical twinning systems (can be used to simulate brass-type textures). "bcc.dat" stands for {110}<111> + {112}<111> slip systems.

line 3: Increasing the two first switches to 2, 3 or 4 will increase the amount of information that is printed out. To be used with care.

line 4: Last line of a section of the input that can be used to modify the values of the critical resolved shear stresses (CRSS). Default value is 1.0 for all of them. In the present example, this section is void except for the last line which must always be present. One may have this line preceded by any number of lines of the following type:

$$n_1 \ n_2 \ \tau^+ \ \tau^-$$

which will replace the default values of the CRSS of systems numbered  $n_1$  through  $n_2$  (both integers) to  $\tau^+$  and  $\tau^-$  (reals) for resp. the positive and the negative slip directions (the program prints the description of the slip systems out; such listing can be used to know the numbers of the slip systems).

line 5: do not touch

lines 6-7: resp.  $\mu$  and  $\Gamma_0$  of section 6.5 of ref. [1]. There is no reason to change this.

line 8: "0" means: the initial texture is in a "TX0-type" file; "1" means: it is in "TX1-type" file. See the additional remarks below.

line 9: name of the file with the initial texture

line 10: An alfanumerical code of max. 4 characters

line 11: Several simulations ("runs") can be performed using the same input files. The lines nr. 15 and following must be repeated for each "run" to be performed.

line 12: Do not change (For advanced users only).

line 13: if "0": no output file; line 14 is not necessary. If "1": an output file will be made. In the latter case, put the name of the output file on the next line, preferably a name that has the suffix .TX1 (see remarks below)

line 15: Title of max. 40 characters

lines 16-17: Describe which texture must be taken from the input file. See remarks below. The values of the example are suitable for TX0 input files.

line 18: "10" means that the same small deformation increment will successively be applied 10 times to each crystallite of the initial texture.

line 19: If "0": a full constraints Taylor model will be used; lines 20-22 are not necessary. If "1": a relaxed Taylor model with 1 relaxation will be used; the tensor  $\mathbf{K}$  that describes this relaxation (see section 3.3 of [1]) must be put on the 3 following lines. In the present example, the tensor given by eq. (14a) of [1] is put; the "lath model" will be used for the simulation.

The parameter can also take higher values  $n$ ; in such case, the next  $3n$  lines should contain the  $n$  tensors  $\mathbf{K}$  that describe the  $n$  relaxations.

line 23: "Delta Epsilon" is  $\Delta E_{eq} = E_{eq} \Delta t$ ; for  $E_{eq}$ , see eq. (2) in [1];  $\Delta t$  is the time increment in one simulation step. To put it shortly,  $\Delta E_{eq}$  is the increment in macroscopic strain during one simulation step.

lines 24-26: the tensor  $\mathbf{K}_0$  that describes the prescribed strain mode (see eqs. (2-3) in [1]). In the example, a rolling operation (plane strain as an approximation) is simulated;  $x_1$  is the rolling direction,  $x_3$  is the rolling plane normal.

In addition to the output file named on line 14, a file with a printout will be generated under the name "taylors.lst". This file has line lengths up to 133 characters. It can be quite lengthy; this file and the output file TAYS.TX1 need each about 900 KBytes for the present example.

### Additional Remarks.

1. TX0-files. The input file used in the example has the format of a "TX0-file". Such ASCII-file contains 1 texture in the form of a discrete set of orientations.

The first line contains the number of crystallite orientations. The second and following lines contain:

- a sequence number (integer)
- a weighing factor (real);
- the three Euler angles  $\phi_1$ ,  $\Phi$ ,  $\phi_2$  in degrees (reals).

A file like this can be made by hand (e.g. to study a single orientation). The file INTXT00.TX0 is a suitable initial texture for simulating rolling textures of cubic metals starting from a random texture. INTXT02.TX0 is a suitable random texture for cubic materials for simulating shear textures ( $x_3$  is still a twofold axis of sample symmetry;  $x_1$ =shear direction;  $x_2$ =shear plane normal).

2. TX1 files. Such file (binary format; "file of double" in TURBO-Pascal) contains **several** textures in the form of discrete sets of orientations. It is made by a program such as TAYLORS or TAYLORW and contains the initial, final and **all** the intermediary deformation textures generated during the simulation. For this reason, the run-number and the step-number (runs and steps of the simulation during which the file was generated) must be specified in order to select one of these textures. Step number 0 yields the initial texture; positive values yield the deformation texture obtained **after** the specified simulation step.

A TX1-file contains much more information than Euler angles and weighing factors.

3. The program TAYLORW. Its usage is similar to that of TAYLORS; the control file is similar but does not contain the lines 6-7.

### 3.3 Extraction of some of the output textures for further processing.

The output file TAYS.TX1 generated by TAYLORS contained 11 textures. It was decided to choose 2 of them for further processing: the deformation textures after the 5th and the 10th deformation step. This was done by issuing the command:

READTXT

The program READTXT (also a Pascal-program) reads the control file READTXT.CTL, which had been prepared beforehand:

```

1    0          LIST CONTROL PARAMETER
2    1          TYPE OF INPUT TEXTURE-FILE(Filename on next line)
3    tays.tx1
4    2          NUMBER OF RUNS TO BE PERFORMED
5    1          Output texture file? If =1, filename on next line:
```

```

6    temp.smt
7    0      M-factor file parameter.If>1,filename on next line:
8    1      CHOSEN RUN
9    5      CHOSEN STEP
10   0      M-factor file parameter.If>1,filename on next line:
11   1      CHOSEN RUN
12   10     CHOSEN STEP
13
14   { If you want Taylor factors: M-factor file=1: unrelaxed M-factors
      =2: relaxed M-factors.

```

The CHOSEN STEP must be the one that identifies the INITIAL ORIENTATION for which the Taylor factor was calculated. For example, in a 1-STEP simulation, CHOSEN STEP must be 0 (1 identifies the final orientation after the step) }

*Explanation:*

line 1: Do not change

line 2: "1" if the input file is a TX1 file.

line 3: Name of input file

line 4: Number of "runs" to be performed by READTXT. Since it has to extract 2 textures from the file, the value must be "2" in the example. Because of this, lines 7-9 have to appear 2 times.

line 5: "1" if a file with extracted textures is wanted, otherwise "0" (then line 6 is absent)

line 6: Name of output file that will contain the extracted textures.

line 7: Ordinary case: 0. Use 1 or 2 if you wants to extract a file with *M*-factors. "1" is for full constraints Taylor factors, and "2" for relaxed constraints Taylor factors. In both cases, 2 extra lines must be inserted after line 7: one with the name of that output file, and one which contains the number of  $\phi_2$ -values in the grid in Euler space: 19 for cubic metals, and 13 for hexagonal close-packed metals.

lines 8-9: Number of the run and the step (of the simulation during which the input file was generated) that specify the texture to be extracted.

lines 10-12: as lines 7-9, for the second "run" of READTXT (second texture to be extracted).

lines 13-14: Comments

**Remark**

The program READTXT can also be used to extract Taylor-factors (*M*-factors) from a TX1-file. The output files are all ASCII-files.

### 3.4 Generation of C-coefficients of the extracted textures.

The file TEMP.SMT that was made by READTXT was copied into the ODF directory. The program SMTODF was used to transform each of the 2 textures in discrete form that it contained into a continuous O.D.F. The latter is the superposition of a number of Gaussian distributions, one for each crystallite orientation and weighed by the weighing factor of these crystallite orientations. The C-coefficients of these 2 O.D.F. were calculated by issuing the command:

```
smtodf1 (for LMAX=22)
```

This activates the procedure SMTODF1.BAT which looks as follows:

```
smtodf smtodef.i01 smtodef.l01 temp.smt create6.b01 create6.b02 c.001 c.002
```

It is also possible to use:

```
smtodf2 (for LMAX=32)
```

which activates SMTODF2.BAT:

```
smtodf smtodef.i02 smtodef.l02 temp.smt create5.b01 create5.b02 c.002 c.002
```

The files "c.001" and "c.002" are the output files containing even + odd C-coefficients. "smtodef.l01" is a printout file. This procedure uses a parameter file "smtodef.i01" which looks as follows:

```
2 2 4 1 IMAG, IDN, IDM, IPR(O: no listing 1: listing)
LMAX 22 7.00
001/5 fcc roll. smtodef l=22
LMAX 22 7.00
001/5 fcc roll. smtodef l=22
```

#### *Explanation:*

Line 1: See table 1 of the manual of "MTMFHM" for IMAG and IDN. For cubic metals, IDM is 4. It is 6 for hexagonal metals.

Line 2: "22" is the maximum degree of the series expansion (leave it as it is); "7.0" is  $\Phi_0$  (see section 5.9 of the manual of "MTMFHM"), the spread of the Gaussian



distribution that will be put on each crystallite orientation. It is not recommended to use values smaller than  $7^\circ$ .

Line 3: The title of the first set of C-coefficients that will be generated.

Note that lines 2 and 3 lines must be repeated for each output texture ("run") that is contained in the output file TEMP.SMT of READTXT.

The resulting files C.001 and C.002 were copied into the O.D.F. directory of the diskette. Fig. 2 shows the O.D.F. that corresponds to C.002.

### **Remark**

The program PASODF with its procedure PASODF1.BAT and its parameter file PASODF.I01 does a similar job as SMTODF. It produces slightly better (=more precise) C-coefficients but needs more calculation time, except for textures with only a few orientations. Fig. 3 shows the O.D.F. that corresponds to C.004 (=a slightly more precise version of C.002).

There is only one version of PASODF1.BAT, for all values of LMAX up to 34. The parameter file PASODF.I01 looks as follows:

```

34 (LMAX THAT CORRESPONDS TO THE INPUT LIBRARIES)
 2 (IMAG)
 1 (IDN)
LMAX 22 7.00
test pasodf L=22

```

Do not change the 1st line. 2nd, 3th line: see table 1 in the manual of MTMFHM. 4th and 5th line: same as 2nd and 3th line of SMTODF.I01.

## **REFERENCES**

1. P. Van Houtte: "A Comprehensive Mathematical Formulation of an Extended Taylor-Bishop-Hill Model Featuring Relaxed Constraints, the Renouard-Wintenberger Theory and a Strain Rate Sensitivity Model", *Textures and Microstructures*, **8-9** (1988) 313-350.
2. L.S. Toth and P. Van Houtte: "Discretization Techniques for Orientation Distribution Functions", *Textures and Microstructures* **19** (1992) 229-244.